

Week 9: Fractions II

JSS1 Mathematics — First Term Lesson Note (Deep-Dive Edition)

Introduction

Building on last week's basics, we now learn how to compare fractions of different sizes and identify fractions that represent the same amount, even when written differently.

Equivalent Fractions

Equivalent fractions have the same value even though they look different. You get them by multiplying (or dividing) the numerator and denominator by the **same number**.

Example: $\frac{1}{2} = \frac{2}{4} = \frac{3}{6} = \frac{4}{8}$ (multiply top and bottom by 2, 3, 4...)

Comparing Fractions

Same denominator: just compare the numerators.

Example: $\frac{3}{7}$ vs $\frac{5}{7} \rightarrow \frac{5}{7}$ is bigger ($5 > 3$)

Different denominators: find the LCM of the denominators (this is called the LCD — Lowest Common Denominator), convert both fractions, then compare.

Example 1: Compare $\frac{3}{5}$ and $\frac{7}{10}$.

LCM of 5 and 10 = 10

$\frac{3}{5} = \frac{6}{10}$ (multiply top and bottom by 2)

$\frac{6}{10}$ vs $\frac{7}{10} \rightarrow \frac{7}{10}$ is bigger

Example 2: Compare $\frac{2}{3}$ and $\frac{3}{4}$.

LCM of 3 and 4 = 12

$\frac{2}{3} = \frac{8}{12}$, $\frac{3}{4} = \frac{9}{12}$

$\frac{8}{12}$ vs $\frac{9}{12} \rightarrow \frac{3}{4}$ is bigger

Ordering Fractions

Example: Arrange $\frac{1}{2}$, $\frac{1}{3}$, $\frac{1}{4}$ from smallest to biggest.

LCM of 2, 3, 4 = 12

$\frac{1}{2} = \frac{6}{12}$, $\frac{1}{3} = \frac{4}{12}$, $\frac{1}{4} = \frac{3}{12}$

Ordering: $\frac{3}{12} < \frac{4}{12} < \frac{6}{12} \rightarrow \frac{1}{4} < \frac{1}{3} < \frac{1}{2}$

Real-Life Application

If one shop sells fabric at $\frac{3}{5}$ of a metre per naira and another at $\frac{7}{10}$ of a metre per naira, converting both to the same denominator (as in Example 1 above) shows exactly which shop offers more fabric for the same price.

Vocabulary

Word	Meaning
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Equivalent	Equal in value, though appearing different
LCD (Lowest Common Denominator)	The LCM of the denominators of two or more fractions

Evaluation Questions

- 1 Write three fractions equivalent to $\frac{2}{5}$.
- 2 Compare $\frac{5}{8}$ and $\frac{3}{4}$ — which is bigger?
- 3 Arrange in ascending order: $\frac{2}{3}$, $\frac{1}{2}$, $\frac{3}{4}$
- 4 Find two fractions equivalent to $\frac{3}{4}$.
- 5 Compare $\frac{7}{12}$ and $\frac{5}{8}$ — which is bigger?
- 6 Arrange in descending order: $\frac{3}{5}$, $\frac{1}{2}$, $\frac{7}{10}$
- 7 Explain what "equivalent fractions" means, using your own example.
- 8 Why do we need to find the LCD before comparing fractions with different denominators?
- 9 Is $\frac{4}{8}$ equivalent to $\frac{1}{2}$? Show your working.
- 10 A student says $\frac{3}{5}$ is bigger than $\frac{2}{3}$ because 3 is bigger than 2. Explain why this reasoning is incorrect, and find the correct answer.